Exp No: 7 Date:	Unconstrained Optimization- Nonlinear Least squares

AIM:

The objective of this lab is to understand and implement optimization techniques for solving nonlinear least squares problems without constraints.

PROCEDURE:

a. Problem Formulation: Define the nonlinear least squares objective function. Specify the data points and the model used.

b. Choose Optimization Library: Select an appropriate optimization library like scipy.optimize.

c. Implement Optimization: Set up the objective function. Add constraints if required.

d. Run Optimization: Execute the optimization algorithm. Monitor convergence and adjust if needed.

e. Post-optimization Analysis: Interpret the results. Compare with initial assumptions and data.

Problem-1:

X	Y
1	2.7
2	7.4
3	20.1
4	54.6

Code:

```
import numpy as np
from scipy.optimize import least_squares

def model(p, x):
    return p[0] *

np.exp(p[1] * x) def

residuals(p, x, y):

    return y - model(p,

x)

x = np.array([0,1,2,3,4])

y =

np.array([1,2.7,7.4,20.1,54.

6]) p0 = [1, 0.5]

result = least_squares(residuals, p0, args=(x, y))

print("Optimized Parameters:", result.x)
```

Output:

Optimized Parameters: [1.00 1.00]

RESULT:

Thus, the Gradient Descent Method was successfully executed and verified using Python.